# PHISHING ANALYSIS REPORT

**Target - https://github.com/rf-peixoto/phishing_pot/blob/main/email/sample-1001.eml**

**Prepared by - Anjali Brahmankar**

**Role – SOC Analyst**

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# Phishing Report

## Executive Summary

Phishing is a cybercrime where attackers masquerade as reputable sources via email, text, or phone to steal sensitive information like passwords, credit card details, and personal data. These fraudulent messages often create a false sense of urgency, urging victims to click malicious links or download attachments, resulting in financial loss or identity theft.

## Background

### How Phishing Works

Phishing attacks typically involve fraudulent communication, often through emails, messages, or websites that appear legitimate. The attacker tricks the victim into clicking a malicious link, downloading an attachment, or providing confidential information. Once the victim responds, the attacker gains access to sensitive data or systems.

### Types of Phishing

1. **Email Phishing:** Mass emails sent to many users posing as legitimate organizations.
2. **Spear Phishing:** Targeted attacks aimed at specific individuals or organizations.
3. **Smishing:** Phishing conducted via SMS or text messages.
4. **Vishing:** Voice-based phishing using phone calls.
5. **Clone Phishing:** A legitimate message is copied and altered with malicious links or attachments.

### Recent trends in phishing

AI phishing: AI phishing uses generative artificial intelligence (AI) tools to create phishing messages. These tools can generate tailored emails and text messages that lack spelling errors, grammatical inconsistencies and other common red flags of phishing attempts.

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Quishing: uses fake QR codes embedded in emails and text messages or posted in the real world. Quishing allows hackers to hide malicious websites and software in plain sight.

Hybrid vishing: These attacks combine voice phishing with other methods to evade spam filters and gain victims' trust.

**Common Techniques Used**

- Creating fake websites that resemble official ones
- Using urgent or alarming messages to pressure victims
- Spoofing email addresses or phone numbers
- Embedding malicious links or attachments

**Impacts of Phishing**

- Financial loss
- Identity theft
- Unauthorized access to accounts
- Data breaches in organizations
- Damage to reputation and trust

**Prevention Measures**

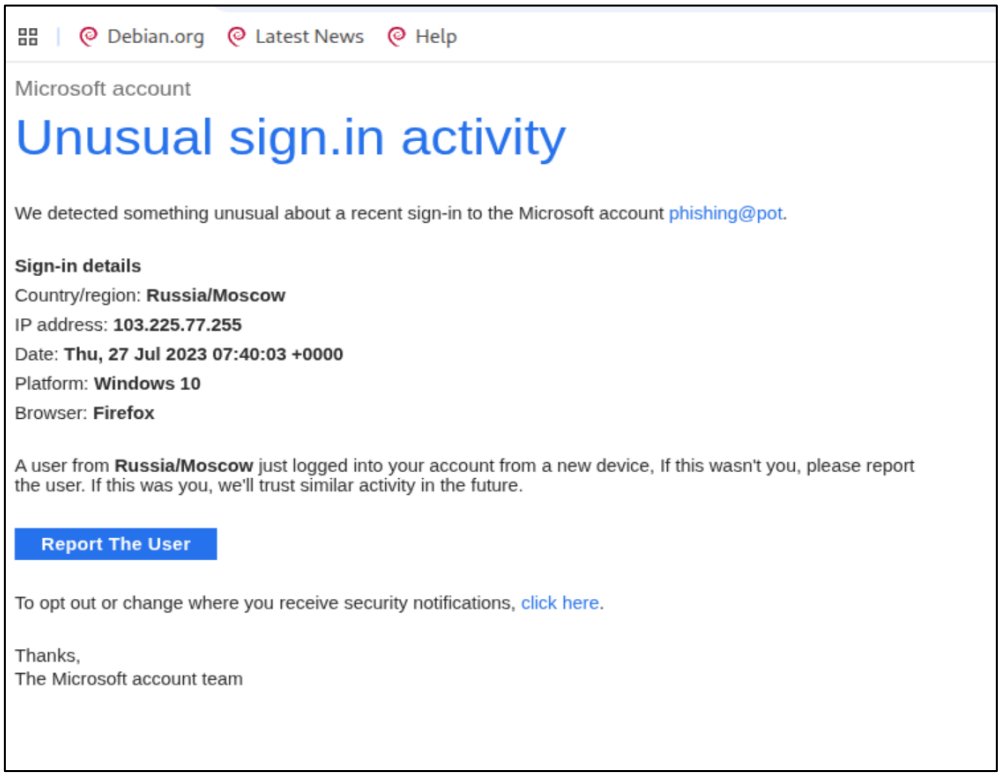
- Verify the sender before responding to messages
- Avoid clicking on suspicious links or attachments
- Use strong, unique passwords and enable two-factor authentication
- Keep software and antivirus updated
- Educate users about cybersecurity awareness

# Phishing Report

## Phishing Analysis

### Sample Email for analysis

Target: [https://github.com/rf-peixoto/phishing_pot/blob/main/email/sample-1001.eml](https://github.com/rf-peixoto/phishing_pot/blob/main/email/sample-1001.eml)



Debian.org Latest News Help

Microsoft account

Unusual sign.in activity

We detected something unusual about a recent sign-in to the Microsoft account phishing@pot.

**Sign-in details**
Country/region: **Russia/Moscow**
IP address: **103.225.77.255**
Date: **Thu, 27 Jul 2023 07:40:03 +0000**
Platform: **Windows 10**
Browser: **Firefox**

A user from **Russia/Moscow** just logged into your account from a new device, If this wasn't you, please report the user. If this was you, we'll trust similar activity in the future.

Report The User

To opt out or change where you receive security notifications, click here.

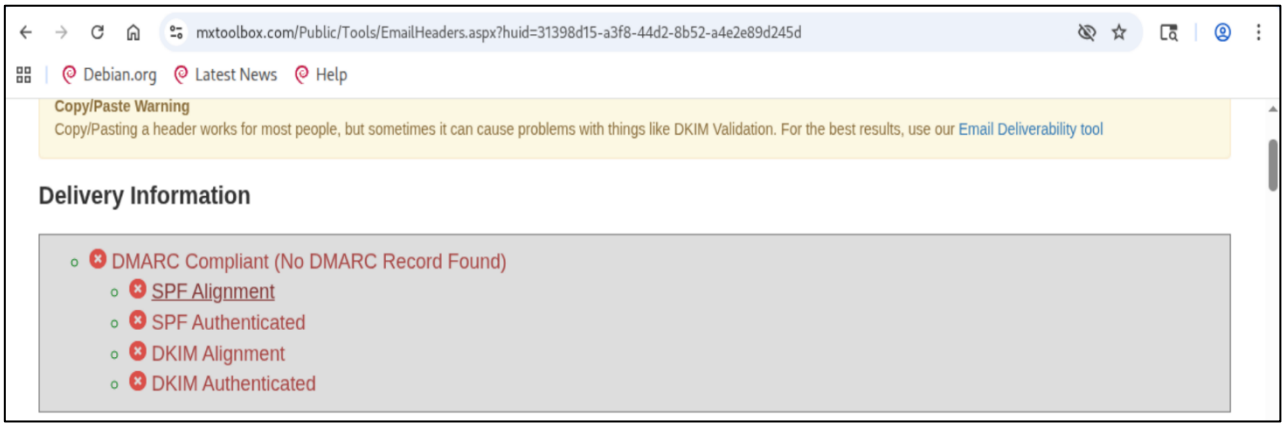
Thanks,
The Microsoft account team

*Figure 1: Sample email 1*

## Analysis Methodology

### Email header analysis

Below are the screenshots of result of email header analysis using MXToolbox



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**Authentication Results:**

The above evidence shows that

1. spf=none, dkim=none, dmarc=permerror, SPF=none
   This means that there's no permission for that server to send email.
   SPF normally lists allowed servers and DMARC uses SPF + DKIM to verify sender. Here, the authentication is broken.
   Conclusion- Strong indicator of spoofing
2. DKIM = none
   This means no digital signature.
   Whereas, legit companies always sign emails
   Conclusion- Message integrity NOT verified
3. There are inconsistencies like:
   "From: access-accsecurity.com", "Return-Path: nonkfrgr.co.uk".
   These should match in legitimate emails. But there's a mismatch.
   Conclusion- spoofing attempt

**Headers Found**

| Header Name | Header Value |
|---|---|
| Authentication-Results | spf=none (sender IP is 89.144.9.87) smtp.mailfrom=nonkfrgr.co.uk; dkim=none (message not signed) header.d=none;dmarc=permerror action=none header.from=access-accsecurity.com; |
| Received-SPF | None (protection.outlook.com: nonkfrgr.co.uk does not designate permitted sender hosts) |
| X-IncomingTopHeaderMarker | OriginalChecksum:581851118E8AB60FB557B9652148A1D2ACF0E25191DB3370DD6E91B6F7AF2E34;UpperCasedChecksum:650D4BED98031B7AE86C487AC7AF30C830E0CD64B4F923A90386050DC65A28C4;SizeAsReceived:396;Count:12 |
| From | Microsoft account team ._.<no-reply@access-accsecurity.com> |
| Subject | Microsoft account unusual signin activity |
| To | phishing@pot |
| Content-Length | 18708448 |
| Date | Thu, 27 Jul 2023 07:40:03 +0000 |
| Reply-To | solutionteamrecognizd02@gmail.com |
| Content-Type | text/html; charset="UTF-8" |
| Content-Transfer-Encoding | 8bit |
| X-IncomingHeaderCount | 12 |
| Message-ID | <412928a2-0aad-4b27-959a-5df404e1f07d@VI1EUR06FT066.eop-eur06.prod.protection.outlook.com> |
| Return-Path | bounce@nonkfrgr.co.uk |
| X-MS-Exchange-Organization-ExpirationStartTime | 27 Jul 2023 07:40:03.4716 (UTC) |
| X-MS-Exchange-Organization-ExpirationStartTimeReason | OriginalSubmit |
| X-MS-Exchange-Organization-ExpirationInterval | 1:00:00:00.0000000 |

# Phishing Report

| | |
|---|---|
| X-MS-Exchange-Organization-AuthSource | VI1EUR06FT066.eop-eur06.prod.protection.outlook.com |
| X-MS-Exchange-Organization-AuthAs | Anonymous |
| X-MS-UserLastLogonTime | 7/27/2023 7:32:58 AM |
| X-MS-Office365-Filtering-Correlation-Id | 92ccd2d7-4aa5-4320-c208-08db8e74b0af |
| X-MS-Exchange-EOPDirect | true |
| X-Sender-IP | 89.144.9.87 |
| X-SID-PRA | NO-REPLY@ACCESS-ACCSECURITY.COM |
| X-SID-Result | NONE |
| X-MS-Exchange-Organization-PCL | 2 |
| X-MS-Exchange-Organization-SCL | 5 |
| X-Microsoft-Antispam | BCL:6; |
| X-MS-Exchange-CrossTenant-OriginalArrivalTime | 27 Jul 2023 07:40:03.3778 (UTC) |
| X-MS-Exchange-CrossTenant-Network-Message-Id | 92ccd2d7-4aa5-4320-c208-08db8e74b0af |
| X-MS-Exchange-CrossTenant-Id | 84df9e7f-e9f6-40af-b435-aaaaaaaaaaaa |
| X-MS-Exchange-CrossTenant-AuthSource | VI1EUR06FT066.eop-eur06.prod.protection.outlook.com |
| X-MS-Exchange-CrossTenant-AuthAs | Anonymous |

4. The" X-MS-Exchange-Organization-SCL: 5" indicates Spam confidence level is high; Microsoft already suspects it.

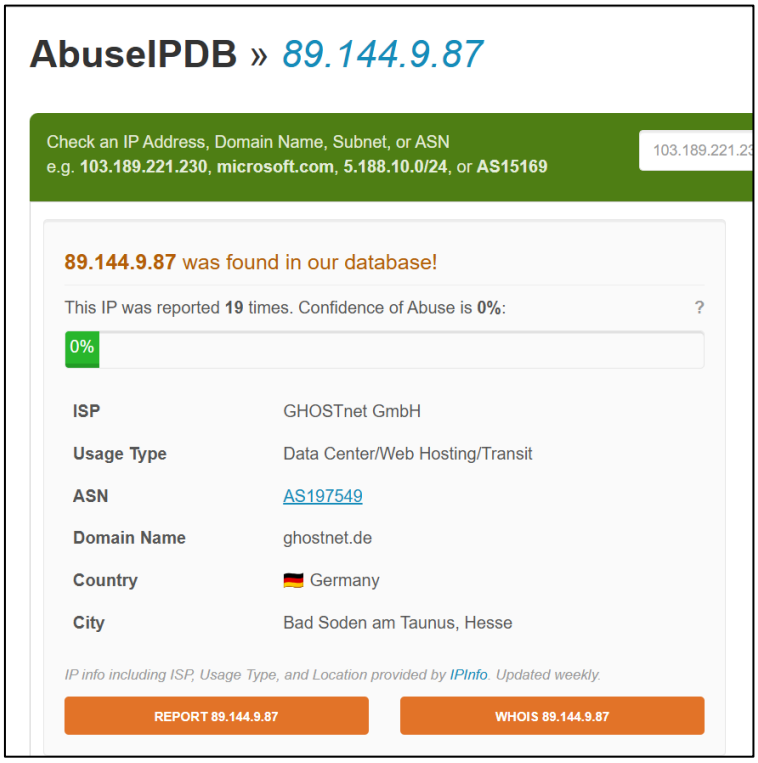
**Headers Found**

| Header Name | Header Value |
|---|---|
| Authentication-Results | spf=none (sender IP is 89.144.9.87) smtp.mailfrom=nonkfrgr.co.uk; dkim=none (message not signed) header.d=none;dmarc=permerror action=none header.from=access-accsecurity.com; |
| Received-SPF | None (protection.outlook.com: nonkfrgr.co.uk does not designate permitted sender hosts) |
| X-IncomingTopHeaderMarker | OriginalChecksum:581851118E8AB60FB557B9652148A1D2ACF0E25191DB3370DD6E91B6F7AF2E34;UpperCasedChecksum:650D4BED98031B7AE86C487AC7AF30C830E0CD64B4F923A90386050DC65A28C4;SizeAsReceived:396;Count:12 |
| From | Microsoft account team ,_<no-reply@access-accsecurity.com> |
| Subject | Microsoft account unusual signin activity |
| To | phishing@pot |
| Content-Length | 18708448 |
| Date | Thu, 27 Jul 2023 07:40:03 +0000 |
| Reply-To | solutionteamrecognizd02@gmail.com |
| Content-Type | text/html; charset="UTF-8" |
| Content-Transfer-Encoding | 8bit |
| X-IncomingHeaderCount | 12 |
| Message-ID | <412928a2-0aad-4b27-959a-5df404e1f07d@VI1EUR06FT066.eop-eur06.prod.protection.outlook.com> |
| Return-Path | bounce@nonkfrgr.co.uk |
| X-MS-Exchange-Organization-ExpirationStartTime | 27 Jul 2023 07:40:03.4716 (UTC) |
| X-MS-Exchange-Organization-ExpirationStartTimeReason | OriginalSubmit |
| X-MS-Exchange-Organization-ExpirationInterval | 1:00:00:00.0000000 |

5. "Reply-To: Gmail address": It claims Microsoft but replies go to Gmail attacker

## IP intelligence test

- IP address verification using AbuseIPDB




The email claims sign-in from Russia while the IP address seems to be from Germany. The country mismatch is an evidence that the email might be a scam.

### Infrastructure Analysis

The domains found - access-accsecurity.com seem to be suspicious, nonkfrgr.co.uk seems to be unrelated.

### Social Engineering Indicators

Urgency ("unusual sign-in"), Fear (foreign location), Fake branding (Microsoft), Malicious response path

## Conclusion

All the above analysis results conclude that the email analysed is a phishing email.

# Phishing Report

## User Awareness Guidelines

End users should adopt the following practices:

- **Verify the sender**: Check the full email address, not just the display name.
- **Avoid clicking embedded links**: Access websites by typing the official URL manually.
- **Inspect links before clicking**: Hover over hyperlinks to confirm legitimacy.
- **Do not open unknown attachments**: avoid opening files from untrusted or unexpected sources.
- **Be cautious of urgency**: Treat time-sensitive requests with skepticism.
- **Never share credentials** via email or messaging platforms.

## Technical Prevention Measures

Organizations should implement layered security controls:

- **Multi-Factor Authentication (MFA)** to protect accounts even if credentials are compromised
- **Email filtering and anti-phishing solutions** to detect and block malicious messages
- **Domain-based protections** such as SPF, DKIM, and DMARC
- **Endpoint protection software** to detect malware from attachments
- **Regular software updates and patch management**

## Organizational Best Practices

- **Security awareness training** conducted regularly for employees
- **Simulated phishing campaigns** to test and improve user resilience
- **Clear reporting procedures** for suspicious emails
- **Least privilege access control** to limit potential damage
- **Verification protocols** for financial or sensitive requests (e.g., callback procedures)

## Incident Response Guidelines

If a phishing attempt is suspected or successful:

1. Do not click any further links or download attachments
2. Report the email to the IT/security team immediately
3. Change compromised passwords promptly
4. Enable or verify MFA settings
5. Scan the system for malware
6. Monitor accounts for unauthorized activity